KAIZEN MATHS

Lesson starter

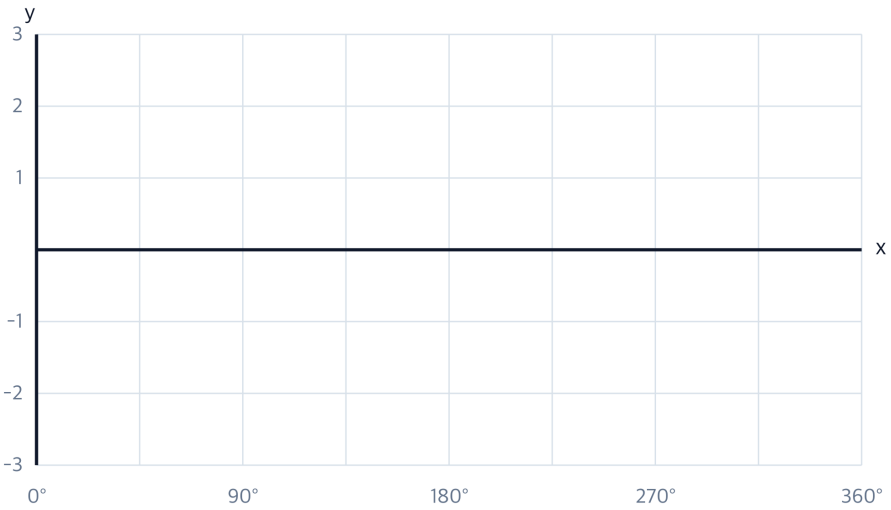
Trigonometric Graphs and Transformations

Standard Graphs

Name _____

SA1

Sketch each standard trigonometric graph and mark the key points.



Sketch y = sin(x) for 0° ≤ x ≤ 360°. Mark the key points.

ALG-020E-L1 · SA

Show your working

KAIZEN MATHS

Lesson starter

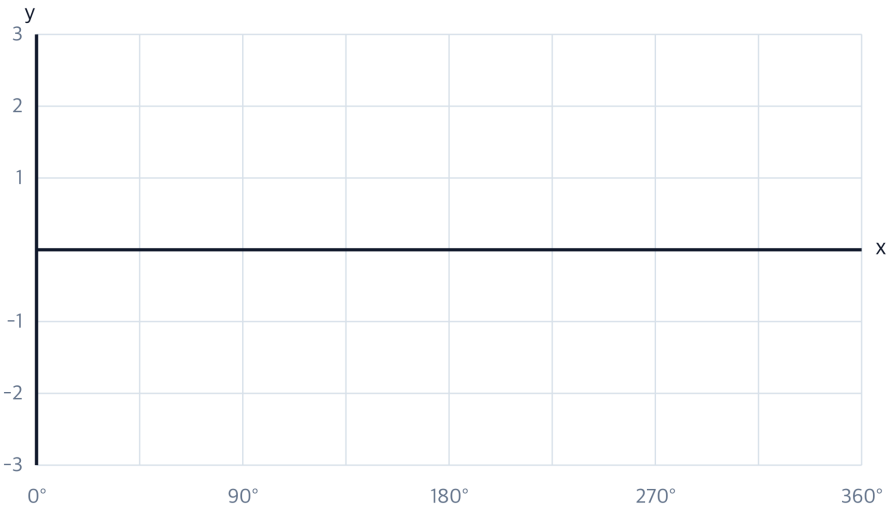
Trigonometric Graphs and Transformations

Standard Graphs

Name _____

SB1

Sketch each tangent graph and mark the asymptotes before drawing the branches.



Sketch y = tan(x) for 0° ≤ x ≤ 360°. Mark the asymptotes.

KAIZEN MATHS

Lesson starter

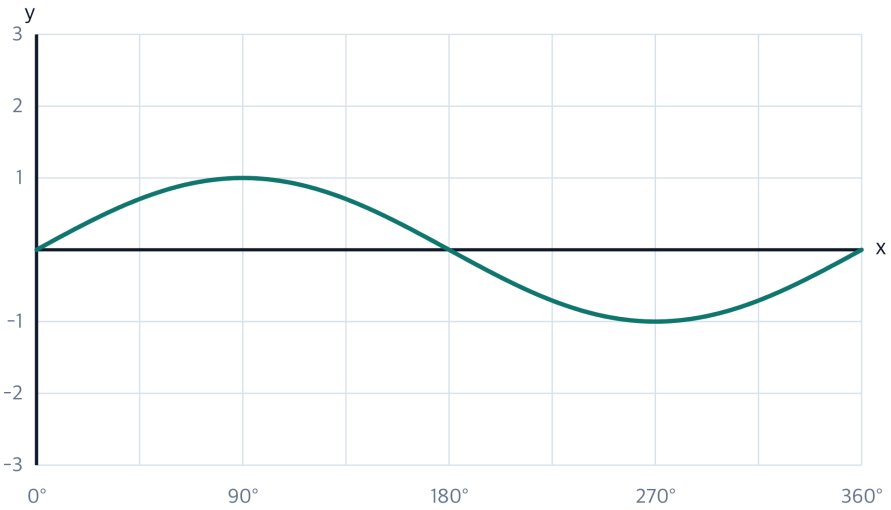
Trigonometric Graphs and Transformations

Standard Graphs

Name _____

SC1

Identify each standard trigonometric graph.



Identify the standard trigonometric graph shown.

ALG-020E-L1 · SC

Show your working

KAIZEN MATHS

Lesson starter

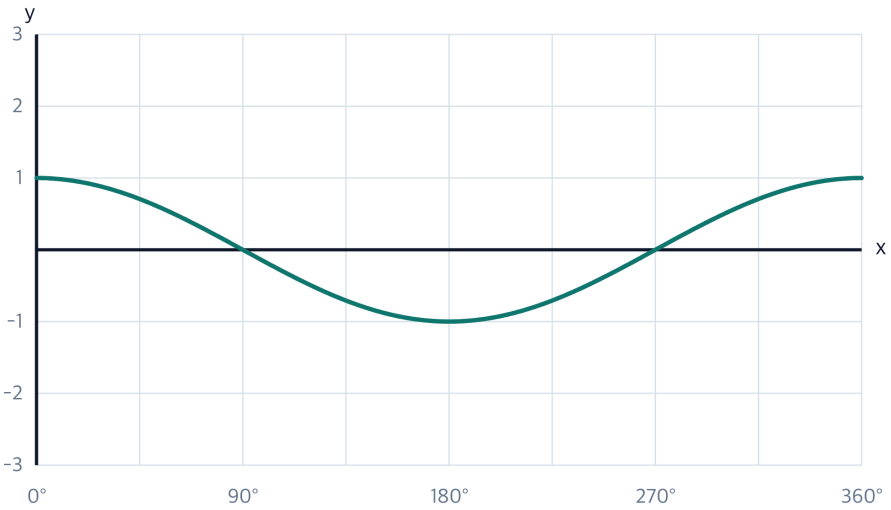
Trigonometric Graphs and Transformations

Standard Graphs

Name _____

SD1

State the amplitude, period, midline, and key graph features.



State the key features of y = cos(x).

ALG-020E-L1 · SD

Show your working

KAIZEN MATHS

Exit ticket

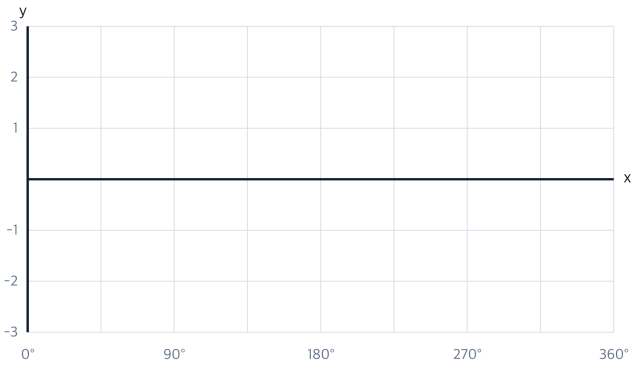
Trigonometric Graphs and Transformations

Standard Graphs

Name _____

EA1

Sketch each standard trigonometric graph and mark the key points.



Sketch y = cos(x) for 0° ≤ x ≤ 360°. Mark the key points.

Show your working

KAIZEN MATHS

Exit ticket

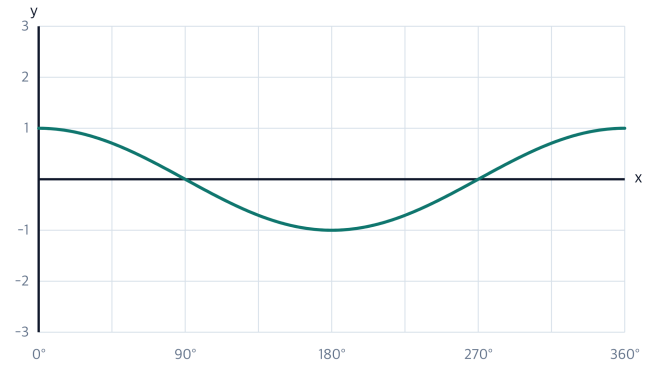
Trigonometric Graphs and Transformations

Standard Graphs

Name _____

EB1

Identify each standard trigonometric graph.



Identify the standard trigonometric graph shown.

Show your working

KAIZEN MATHS

Exit ticket

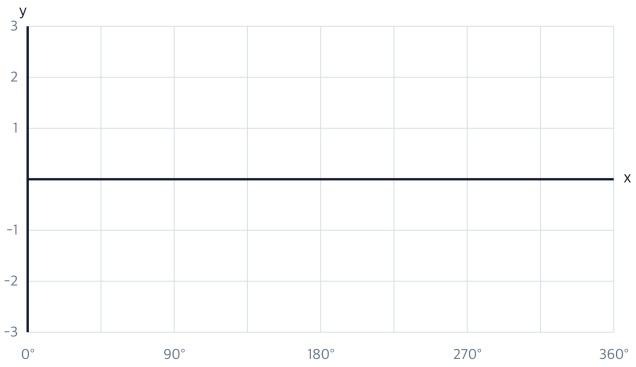
Trigonometric Graphs and Transformations

Standard Graphs

Name _____

EA1

Sketch each standard trigonometric graph and mark the key points.



Sketch y = cos(x) for 0° ≤ x ≤ 360°. Mark the key points.

ALG-020E-L1 · EA

Show your working

KAIZEN MATHS

Exit ticket

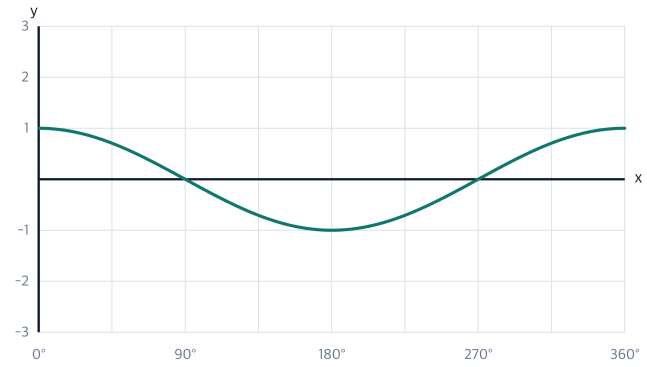
Trigonometric Graphs and Transformations

Standard Graphs

Name _____

EB1

Identify each standard trigonometric graph.



Identify the standard trigonometric graph shown.

ALG-020E-L1 · EB

Show your working

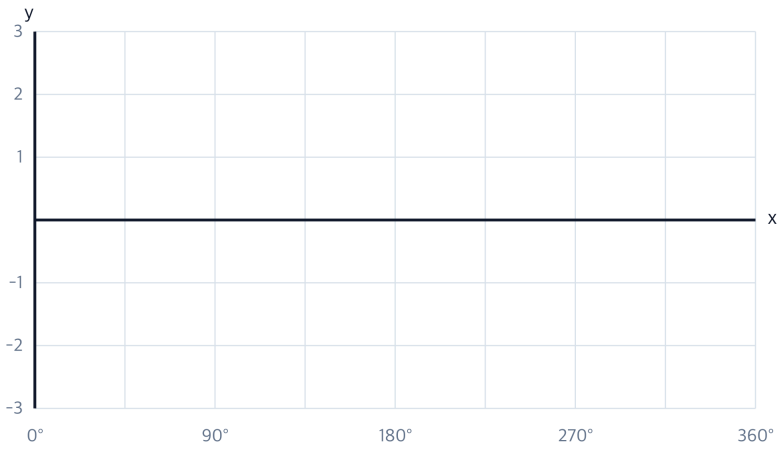
Teacher answers & working

Trigonometric Graphs and Transformations · Standard Graphs

ALG-020E-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

Sketch each standard trigonometric graph and mark the key points.



Sketch y = sin(x) for 0° ≤ x ≤ 360°. Mark the key points.

Answer:

Sine graph with amplitude 1 and period 360°.

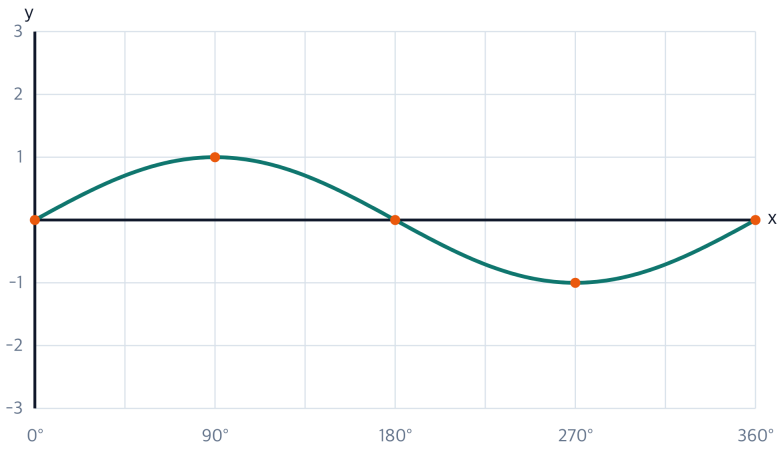
Working:

Use the standard sin graph over one full cycle.

The amplitude is 1, the midline is y = 0, and the period is 360°.

Plot the key points.

x	y
0°	0
90°	1
180°	0
270°	-1
360°	0



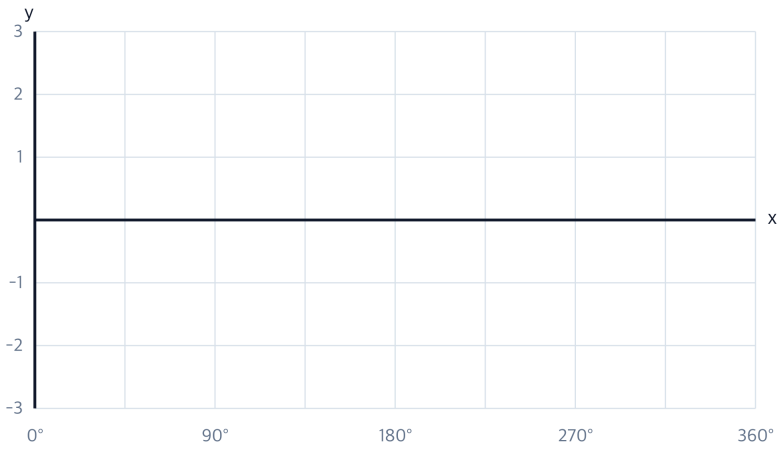
Teacher answers & working

Trigonometric Graphs and Transformations · Standard Graphs

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SB1

Sketch each tangent graph and mark the asymptotes before drawing the branches.



Sketch y = tan(x) for 0° ≤ x ≤ 360°. Mark the asymptotes.

Answer:

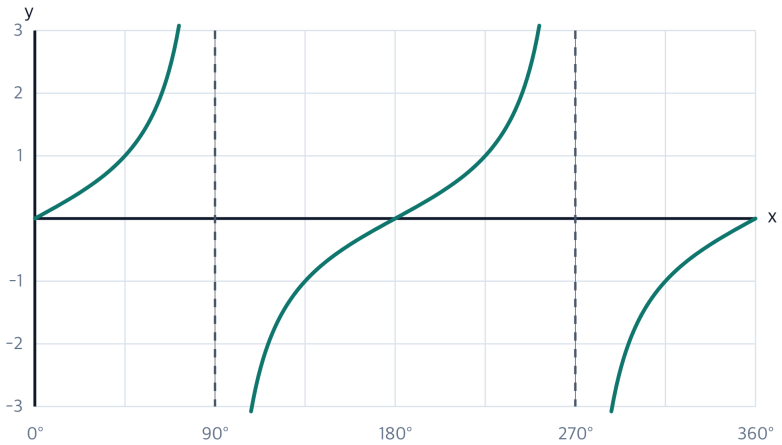
Tangent graph with period 180° and asymptotes x = 90° and x = 270°.

Working:

The tangent graph has period 180°.

The graph crosses the x-axis at 0°, 180°, and 360°.

The vertical asymptotes occur at 90° and 270°.



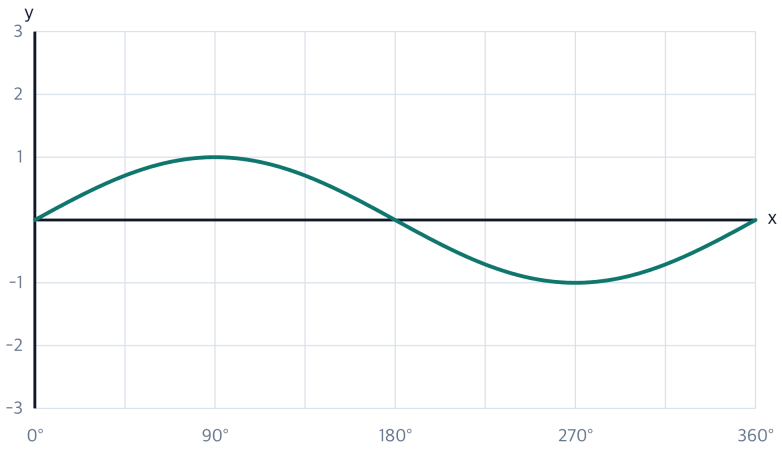
Teacher answers & working

Trigonometric Graphs and Transformations · Standard Graphs

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SC1

Identify each standard trigonometric graph.



Identify the standard trigonometric graph shown.

Answer:

$y = \sin x$.

Working:

Look at where the graph begins and whether it has asymptotes.

The graph starts at (0°, 0), rises to 1, returns to 0, falls to -1, and returns to 0.

Therefore the graph is y = sin x.

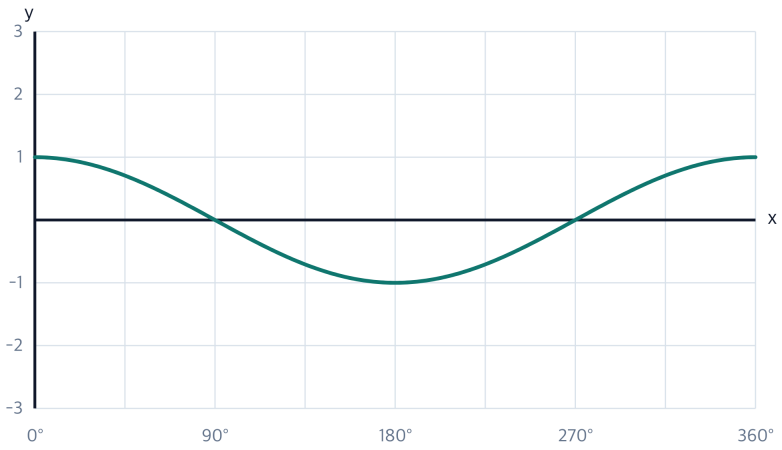
Teacher answers & working

Trigonometric Graphs and Transformations · Standard Graphs

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SD1

State the amplitude, period, midline, and key graph features.



State the key features of y = cos(x).

Answer:

Amplitude 1, period 360°, midline y = 0.

Working:

Start from the parent graph y = cos x.

The coefficient outside the trigonometric function is 1, so the amplitude is |1| = 1.

The coefficient inside the bracket is 1, so the period is 360° ÷ 1 = 360°.

The graph is shifted 0° horizontally.

The + 0 outside the function gives the midline y = 0.

Summarise the graph features.

Amplitude	1
Period	360°
Midline	$y = 0$
Phase shift	none

Use the transformed key points to guide the sketch.

x	y
0°	1
90°	0
180°	-1
270°	0
360°	1

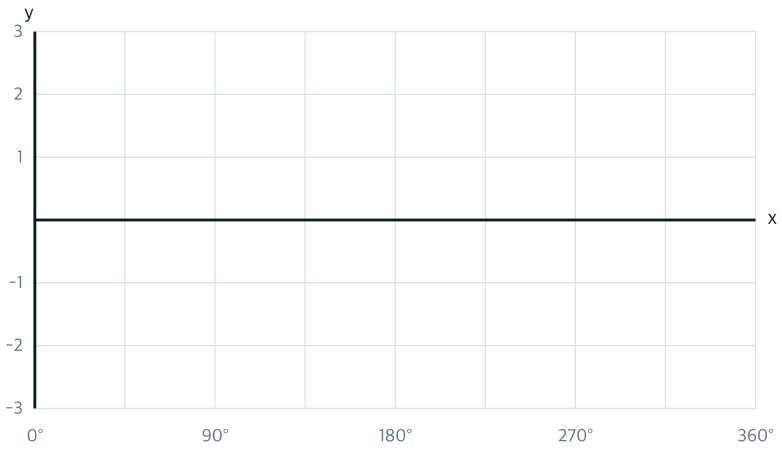
Teacher answers & working

Trigonometric Graphs and Transformations · Standard Graphs

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EA1

Sketch each standard trigonometric graph and mark the key points.



Sketch y = cos(x) for 0° ≤ x ≤ 360°. Mark the key points.

Answer:

Cosine graph with amplitude 1 and period 360°.

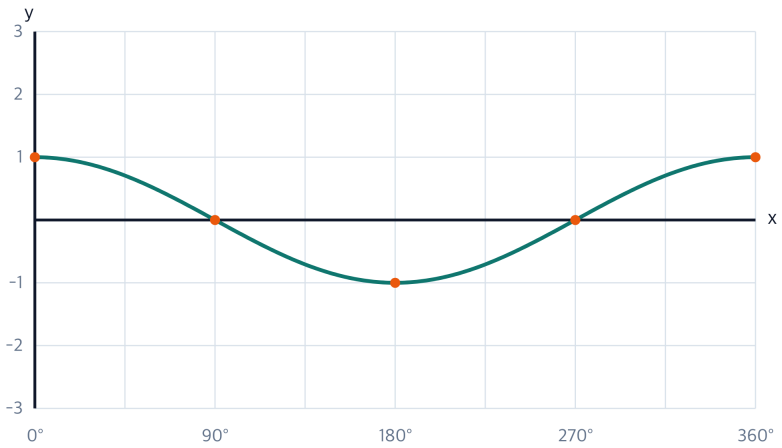
Working:

Use the standard cos graph over one full cycle.

The amplitude is 1, the midline is y = 0, and the period is 360°.

Plot the key points.

x	y
0°	1
90°	0
180°	-1
270°	0
360°	1



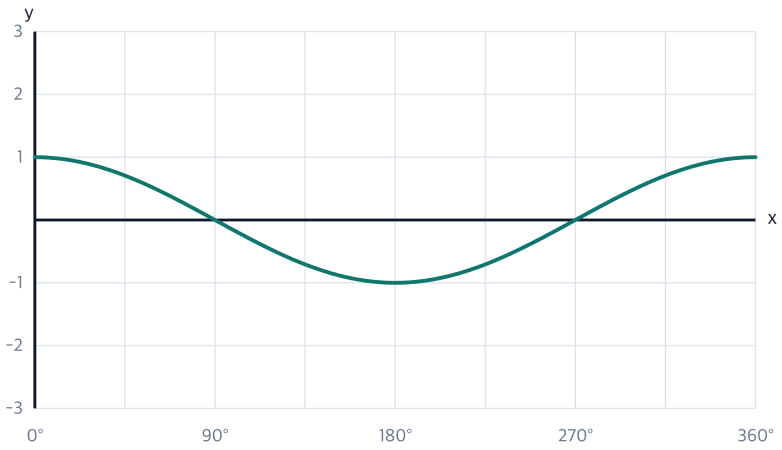
Teacher answers & working

Trigonometric Graphs and Transformations · Standard Graphs

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EB1

Identify each standard trigonometric graph.



Identify the standard trigonometric graph shown.

Answer:

$y = \cos x$.

Working:

Look at where the graph begins and whether it has asymptotes.

The graph starts at (0°, 1), falls to -1 at 180°, then returns to 1.

Therefore the graph is y = cos x.